

Lab 12 – A Simple Code Generator

Aim

To generate simple assembly-like target code from an arithmetic expression.

Algorithm

1. Start the program.
2. Read an arithmetic expression of the form $x = y \text{ op } z$.
3. Store:
 1. destination variable
 2. first operand
 3. operator
 4. second operand
4. Generate target code:
 1. MOV R0, operand1
 2. perform operation with operand2
 3. store result in destination
5. Display the generated code.
6. Stop.

C Program

```
#include <stdio.h>

int main() {
    char expr[20];
    char dest, op1, op2, opr;

    printf("Enter expression (Example: a=b+c): ");
    scanf("%s", expr);

    dest = expr[0];
    op1 = expr[2];
    opr = expr[3];
    op2 = expr[4];

    printf("\nGenerated Target Code:\n");

    printf("MOV R0, %c\n", op1);

    switch (opr) {
        case '+':
            printf("ADD R0, %c\n", op2);
            break;
    }
```

```

    case '-':
        printf("SUB R0, %c\n", op2);
        break;

    case '*':
        printf("MUL R0, %c\n", op2);
        break;

    case '/':
        printf("DIV R0, %c\n", op2);
        break;

    default:
        printf("Invalid operator\n");
        return 0;
}

printf("MOV %c, R0\n", dest);

return 0;
}

```

Sample Input

a=b+c

Sample Output

Generated Target Code:
MOV R0, b
ADD R0, c
MOV a, R0

Result

Thus a simple code generator was implemented successfully to generate target code for the given arithmetic expression.